

Abstract

This report presents the design and analysis of a complete electrical circuit in the home that incorporates three bedrooms, a kitchen, a living room, a guest room, a garage and a lounge along with an integrated Uninterrupted Power Supply (UPS) system. The study includes the selection of electrical loads based on typical household requirements and the identification of appliances suitable for UPS support, restricted to ratings below 100W. Detailed calculations of current, voltage, and power are performed for individual components, each room, and the overall home system for both main supply and UPS operation. MATLAB is used to perform the analytical computations, while the circuit is simulated using Multi-sim to verify performance and ensure proper component selection. The report demonstrates an engineering approach to designing a practical residential wiring system with reliable backup power, highlighting the importance of load management, efficiency, and safety in home electrical networks.

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Chapter 1

Introduction

Electrical power distribution in residential buildings plays a critical role in ensuring safe, reliable, and efficient operation of household appliances. A typical home comprises multiple rooms with diverse electrical load requirements, such as lighting, fans, sockets, and essential low-power devices. To maintain continuity of supply during power outages, an Uninterrupted Power Supply (UPS) system is often integrated into the wiring network. Designing such a circuit requires a thorough understanding of basic electrical principles, accurate load estimation, component selection, and practical implementation standards.

This project focuses on developing a complete electrical circuit for a home consisting of three bedrooms, a kitchen, a living room, and a guest room. The design includes proper load distribution, protection elements, and a dedicated UPS connection for appliances rated below 100W. The analysis involves calculating current, voltage, and power for different parts of the system, followed by comparative evaluation under main supply and UPS operation. MATLAB is used to perform the required computations, while circuit simulations are carried out in Multisim to validate the design. This work aims to strengthen the understanding of residential wiring concepts and the application of modern engineering tools in solving practical circuit-related problems.

1.1 Objectives

The Objectives of this Complex Engineering Problem Are...

1. To design a complete electrical circuit for a home containing three bedrooms, a kitchen, a living room, and a guest room.
2. To select appropriate electrical loads and identify appliances suitable for UPS operation (below 100W).
3. To calculate current, voltage, and power for each component, each room, and the overall home system.
4. To compare circuit performance under both main supply and UPS operation.
5. To choose suitable component ratings ensuring safety, efficiency, and proper load management.
6. To use MATLAB for performing all analytical calculations.

7. To simulate the designed circuit using Multi-sim for validation.
8. To prepare a structured technical report based on the design, calculations, and simulations..

1.2 House Plan and Abbreviations

1.2.1 Floor Plan of House

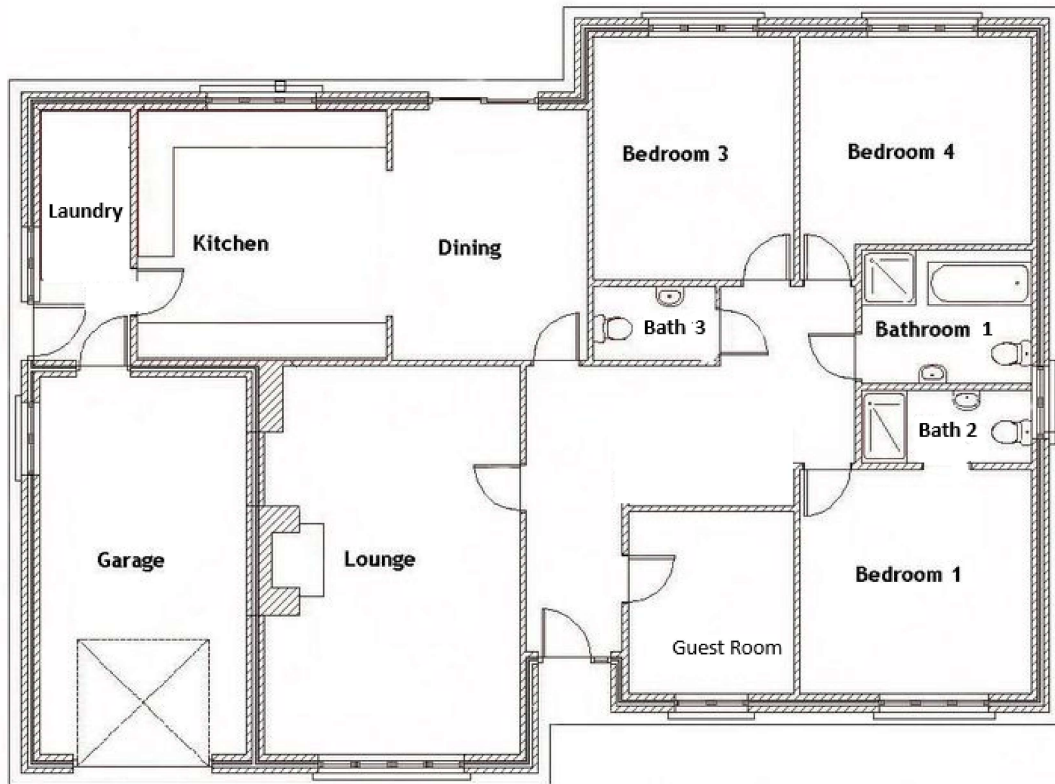


Figure 1.1: Floor Plan of the House

1.2.2 Rooms and There Abbreviations

We use these room abbreviations to make our MATLAB calculations easier and more organized. Instead of typing long room names repeatedly, the abbreviations let us reference each room quickly. This also keeps the code cleaner and reduces the chances of errors.

Table 1.1: Room Abbreviations

Room	Abbreviation
Bed Room 1	B1
Bed Room 2	B2
Bed Room 3	B3
Guest Room	G
Bath Room 1	Bath1
Bath Room 2	Bath2
Bath 3 and Hall	Bath3nH
Dining Room	D
Lounge	L
Kitchen	K
Laundry	Ly
Garage	Ge

1.3 Appliances and there Power Rating

Table 1.2: Appliances and there Power Rating

Name	Power	Abbreviation
Fan	50W	Fan
Bulb	15W	Bulb
2 ton AC	2000W	2ton AC
1 ton AC	1300W	1ton AC
Computer	200W	Comp
Exhaust	25W	Ext
TV	180W	TV
Iron	1500W	Iron
Oven	1500W	Oven
Refrigerator	200W	Refg
Washing Machine	400W	WashMach
Dryer	1000W	Dryer
Water Pump	1000W	Wpump
Low Power Plug	100W	LP
High Power Plug	1000W	HP

Chapter 2

Multisim Simulation Of Ordinary Home

2.1 Software

For the design and analysis of our electrical circuit, we used NI Multisim 14.3, a professional simulation software widely employed in electronics engineering. Multisim allowed us to create a complete schematic by placing components such as power sources, switches, loads, and wiring according to our circuit requirements. The software also enabled us to apply our color-coding scheme—red for high-power lines, green for low-power circuits, black for neutral, and blue for grounding—to clearly distinguish different sections of the design.

Using Multisim’s simulation tools, we tested the behavior of the circuit under various conditions, observed voltage and current flow, and ensured proper functioning of the grounding and protective paths. This helped us verify the circuit’s reliability and safety before any real-world implementation. Overall, Multisim provided an accurate, efficient, and user-friendly platform that improved both the design quality and understanding of the system.

2.2 Circuit Diagram

The following diagram illustrates the complete electrical wiring layout of a residential house that includes three bedrooms, a guest room, a lounge, kitchen, dining room, laundry area, garage, and multiple bathrooms. Each room is represented with its relevant loads, switches, and protective components to reflect a realistic and organized electrical system. This layout demonstrates how power is distributed across the entire house while maintaining proper load management and safety measures.

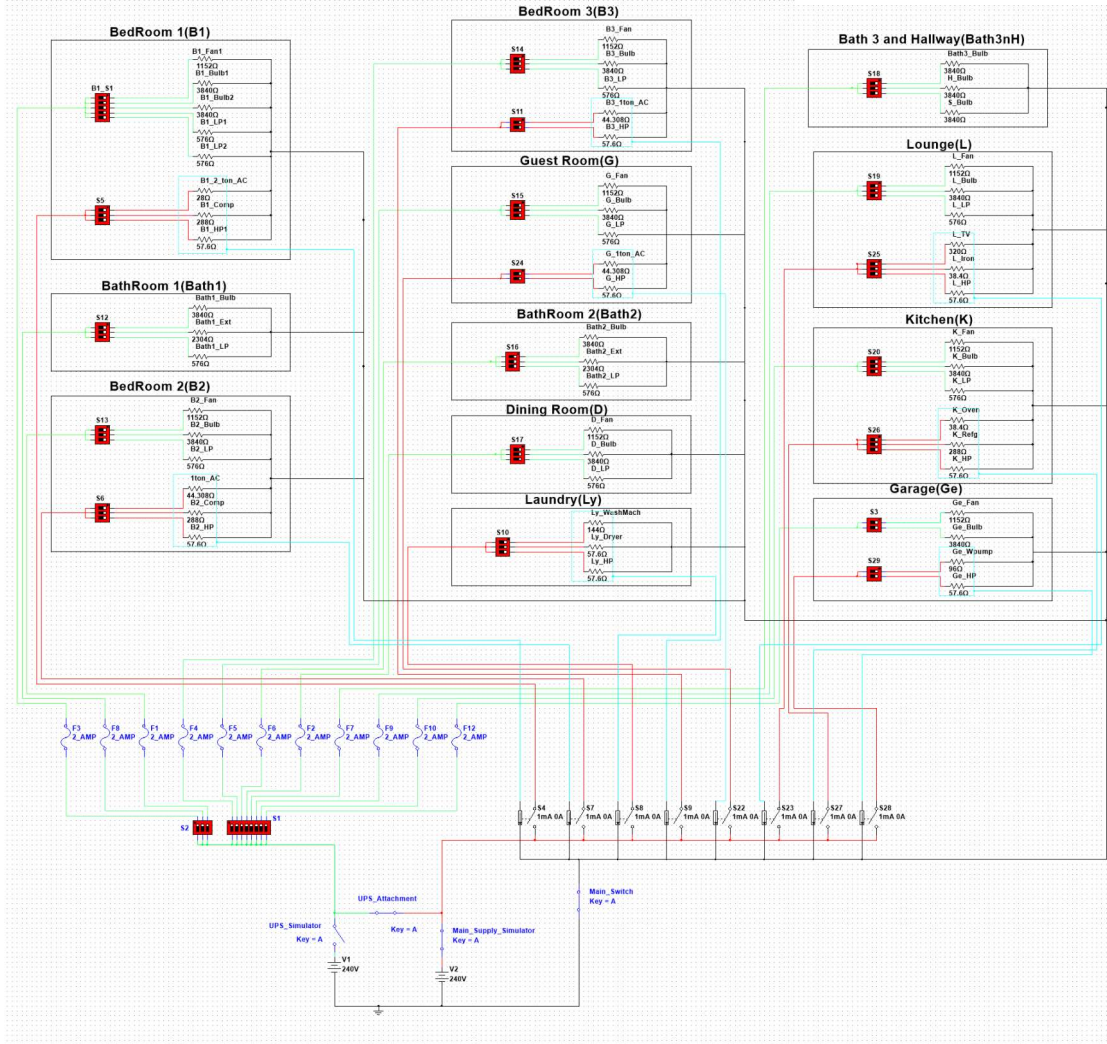


Figure 2.1: Circuit Diagram of home comprising three bedrooms, a kitchen, a living room, and a guest room.

2.3 Wires Color Coding

2.3.1 High-Power Circuits (Red Wires)

The red wires represent the high-power supply lines. These deliver current to heavy-load appliances such as air conditioners, ovens, washing machines, and other equipment that requires higher wattage. Marking these circuits in red makes it easy to identify high-load paths for proper breaker sizing and safer installation.

2.3.2 Low-Power Circuits (Green Wires)

The green wires indicate low-power circuits, typically used for lighting, fans, switches, sensors, and other light-duty loads. Since these circuits draw much less current, separating them with a unique color helps reduce confusion and makes troubleshooting simpler.

2.3.3 Neutral Connections (Black Wires)

The black wires are used as the neutral conductors. They provide the return path for electric current back to the main panel, completing the circuit. Using black for neutral helps clearly distinguish it from both high-power and low-power lines, reducing wiring mistakes.

2.3.4 Earth Connections (Blue Wires)

The blue wires represent the ground (earth). These wires are connected to the metal cases, conduits, and all exposed conductive parts. In the event of a short circuit or fault, the blue ground wire provides a safe, low-resistance path to earth, ensuring that protective devices trip immediately to prevent electric shock or fire hazards.

2.4 Protection Devices Used in the System

2.4.1 Fuses

In our system, we used fuses to protect the low-power circuits such as lights, fans, and small electronic loads. The fuse contains a thin wire that melts when too much current flows through it. When this happens, the fuse “blows” and cuts off the supply, preventing damage or overheating.

We used fuses because they are simple, inexpensive, and suitable for circuits that draw only small amounts of current.

2.4.2 Current-Controlled Switching Devices

For the high-power circuits, we used current-controlled switching devices such as circuit breakers and ground-fault protection devices. These devices automatically open the circuit when the current becomes too high or when current leaks to the ground.

We used them because high-power equipment requires stronger and more reliable protection, and these devices can detect overloads, short circuits, and ground faults without needing to be replaced like a fuse.

2.5 Assumptions

We use some assumptions in the circuit for our convenience. Which are as follows:

- We assumed a **240 V DC supply** instead of an AC source to simplify the analysis. In reality, most household circuits are AC, and AC calculations involve RMS values, phase angles, and power factor, which can make calculations more complex.
- We assumed each **low-power plug** to be rated at **100 W**, representing small appliances like lights and fans. In practice, these devices often draw less than their rated power, and the wiring resistance further reduces the actual power delivered.
- We assumed each **high-power plug** to be rated at **1000 W**, representing larger appliances such as heaters or motors. The real power consumed by these devices is usually lower than the rated value due to voltage drops and efficiency losses in the wiring.
- For calculation purposes, **we assumed that every component operates at its full rated power**. In real-life conditions, components rarely consume full power continuously, and the conductor resistance and other losses reduce the actual power supplied to the device.

Chapter 3

Calculations & Result

3.1 Software

In this CEP, we use this MATLAB code to calculate the current and resistance of each electrical component based on its power rating and voltage. We use it to separate loads into UPS-supported loads ($\leq 100\text{W}$) and main-supply loads ($> 100\text{W}$). We also use it to keep track of total power and current in both categories, which helps us size the UPS and understand the overall load of the home. This way, we use MATLAB as a modern tool to perform accurate calculations for our circuit design.

3.2 Matlab Code

```
num = input('Number of Components: ');
V = input('V (Volts) = ');
L_P_sum = 0;
H_P_sum = 0;
L_I_sum = 0;
H_I_sum = 0;
fprintf('-----\n');
for i=1:num
    Name = input('Name of Component: ', 's');
    P = input('P (Watts) = ');
    R = V^2 / P;
    I = V / R;
    fprintf('I_s = %.3f A\n', Name, I);
    fprintf('R_s = %.3f Ohms\n', Name, R);
    fprintf('-----\n');
    if P>100
        H_P_sum = H_P_sum + P;
        H_I_sum = H_I_sum + I;
    else
        L_P_sum = L_P_sum + P;
        L_I_sum = L_I_sum + I;
    end
end
```

3.3 Individual Calculations

3.3.1 For Bed Room 1

```
>> CEP
Number of Components :55
V (Volts) = 240
-----
Name of Component: B1_Fan
P(Watts) = 50
I_B1_Fan = 0.208 A
R_B1_Fan = 1152.000  $\Omega$ 
-----
Name of Component: B1_Bulb1
P(Watts) = 15
I_B1_Bulb1 = 0.062 A
R_B1_Bulb1 = 3840.000  $\Omega$ 
-----
Name of Component: B1_Bulb2
P(Watts) = 15
I_B1_Bulb2 = 0.062 A
R_B1_Bulb2 = 3840.000  $\Omega$ 
-----
Name of Component: B1_LP1
P(Watts) = 100
I_B1_LP1 = 0.417 A
R_B1_LP1 = 576.000  $\Omega$ 
-----
Name of Component: B1_2ton_AC
P(Watts) = 2000
I_B1_2ton_AC = 8.333 A
R_B1_2ton_AC = 28.800  $\Omega$ 
-----
Name of Component: B1_Comp
P(Watts) = 200
I_B1_Comp = 0.833 A
R_B1_Comp = 288.000  $\Omega$ 
-----
Name of Component: B1_HP1
P(Watts) = 1000
I_B1_HP1 = 4.167 A
R_B1_HP1 = 57.600  $\Omega$ 
-----
```

Figure 3.1: Bed Room 1 Calculations

3.3.2 For Bed Room 2

```
-----  
Name of Component: B2_Fan  
P(Watts) = 50  
I_B2_Fan = 0.208 A  
R_B2_Fan = 1152.000  $\Omega$   
-----  
Name of Component: B2_Bulb  
P(Watts) = 15  
I_B2_Bulb = 0.062 A  
R_B2_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: B2_LP  
P(Watts) = 100  
I_B2_LP = 0.417 A  
R_B2_LP = 576.000  $\Omega$   
-----  
Name of Component: B2_1ton_AC  
P(Watts) = 1300  
I_B2_1ton_AC = 5.417 A  
R_B2_1ton_AC = 44.308  $\Omega$   
-----  
Name of Component: B2_HP  
P(Watts) = 1000  
I_B2_HP = 4.167 A  
R_B2_HP = 57.600  $\Omega$   
-----  
Name of Component: B2_Comp  
P(Watts) = 200  
I_B2_Comp = 0.833 A  
R_B2_Comp = 288.000  $\Omega$   
-----
```

Figure 3.2: Bed Room 2 Calculations

3.3.3 For Bed Room 3

```
-----  
Name of Component: B3_Fan  
P(Watts) = 50  
I_B3_Fan = 0.208 A  
R_B3_Fan = 1152.000  $\Omega$   
-----  
Name of Component: B3_Bulb  
P(Watts) = 15  
I_B3_Bulb = 0.062 A  
R_B3_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: B3_LP  
P(Watts) = 100  
I_B3_LP = 0.417 A  
R_B3_LP = 576.000  $\Omega$   
-----  
Name of Component: B3_lton_AC  
P(Watts) = 1300  
I_B3_lton_AC = 5.417 A  
R_B3_lton_AC = 44.308  $\Omega$   
-----  
Name of Component: B3_HP  
P(Watts) = 1000  
I_B3_HP = 4.167 A  
R_B3_HP = 57.600  $\Omega$   
-----
```

Figure 3.3: Bed Room 3 Calculations

3.3.4 For Guest Room

```
-----  
Name of Component: G_Fan  
P(Watts) = 50  
I_G_Fan = 0.208 A  
R_G_Fan = 1152.000  $\Omega$   
-----  
Name of Component: G_Bulb  
P(Watts) = 15  
I_G_Bulb = 0.062 A  
R_G_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: G_HP  
P(Watts) = 100  
I_G_HP = 0.417 A  
R_G_HP = 576.000  $\Omega$   
-----  
Name of Component: G_lton_AC  
P(Watts) = 1300  
I_G_lton_AC = 5.417 A  
R_G_lton_AC = 44.308  $\Omega$   
-----  
Name of Component: G_HP  
P(Watts) = 1000  
I_G_HP = 4.167 A  
R_G_HP = 57.600  $\Omega$   
-----
```

Figure 3.4: Guest Room Calculations

3.3.5 For Bath Room 1

```
-----  
Name of Component: Bath1_Bulb  
P(Watts) = 15  
I_Bath1_Bulb = 0.062 A  
R_Bath1_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: Bath1_Ext  
P(Watts) = 25  
I_Bath1_Ext = 0.104 A  
R_Bath1_Ext = 2304.000  $\Omega$   
-----  
Name of Component: Bath1_LP  
P(Watts) = 100  
I_Bath1_LP = 0.417 A  
R_Bath1_LP = 576.000  $\Omega$   
-----
```

Figure 3.5: Bath Room 1 Calculations

3.3.6 For Bath Room 2

```
-----  
Name of Component: Bath2_Bulb  
P(Watts) = 15  
I_Bath2_Bulb = 0.062 A  
R_Bath2_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: Bath2_Ext  
P(Watts) = 25  
I_Bath2_Ext = 0.104 A  
R_Bath2_Ext = 2304.000  $\Omega$   
-----  
Name of Component: Bath2_LP  
P(Watts) = 100  
I_Bath2_LP = 0.417 A  
R_Bath2_LP = 576.000  $\Omega$   
-----
```

Figure 3.6: Bath Room 2 Calculations

3.3.7 For Bath Room 3 and Hallway

```
-----  
Name of Component: B3_Bulb  
P(Watts) = 15  
I_B3_Bulb = 0.062 A  
R_B3_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: H_Bulb  
P(Watts) = 15  
I_H_Bulb = 0.062 A  
R_H_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: S_Bulb  
P(Watts) = 15  
I_S_Bulb = 0.062 A  
R_S_Bulb = 3840.000  $\Omega$   
-----
```

Figure 3.7: Bath Room 3 and Hallway Calculations

3.3.8 For Dining Room

```
-----  
Name of Component: D_Fan  
P(Watts) = 50  
I_D_Fan = 0.208 A  
R_D_Fan = 1152.000  $\Omega$   
-----  
Name of Component: D_Bulb  
P(Watts) = 15  
I_D_Bulb = 0.062 A  
R_D_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: D_LP  
P(Watts) = 100  
I_D_LP = 0.417 A  
R_D_LP = 576.000  $\Omega$   
-----
```

Figure 3.8: Dining Room Calculations

3.3.9 For Lounge

```
-----  
Name of Component: L_Fan  
P(Watts) = 50  
I_L_Fan = 0.208 A  
R_L_Fan = 1152.000  $\Omega$   
-----  
Name of Component: L_Bulb  
P(Watts) = 15  
I_L_Bulb = 0.062 A  
R_L_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: L_LP  
P(Watts) = 100  
I_L_LP = 0.417 A  
R_L_LP = 576.000  $\Omega$   
-----  
Name of Component: L_TV  
P(Watts) = 180  
I_L_TV = 0.750 A  
R_L_TV = 320.000  $\Omega$   
-----  
Name of Component: L_Iron  
P(Watts) = 1500  
I_L_Iron = 6.250 A  
R_L_Iron = 38.400  $\Omega$   
-----
```

Figure 3.9: Lounge Calculations

3.3.10 For Kitchen

```
-----  
Name of Component: K_Bulb  
P(Watts) = 15  
I_K_Bulb = 0.062 A  
R_K_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: K_Fan  
P(Watts) = 50  
I_K_Fan = 0.208 A  
R_K_Fan = 1152.000  $\Omega$   
-----  
Name of Component: K_LP  
P(Watts) = 100  
I_K_LP = 0.417 A  
R_K_LP = 576.000  $\Omega$   
-----  
Name of Component: K_Oven  
P(Watts) = 1500  
I_K_Oven = 6.250 A  
R_K_Oven = 38.400  $\Omega$   
-----  
Name of Component: K_Refg  
P(Watts) = 200  
I_K_Refg = 0.833 A  
R_K_Refg = 288.000  $\Omega$   
-----  
Name of Component: K_Refg  
P(Watts) = 1000  
I_K_Refg = 4.167 A  
R_K_Refg = 57.600  $\Omega$   
-----
```

Figure 3.10: Kitchen Calculations

3.3.11 For Laundry

```
-----  
Name of Component: Ly_WashMach  
P(Watts) = 400  
I_Ly_WashMach = 1.667 A  
R_Ly_WashMach = 144.000  $\Omega$   
  
-----  
Name of Component: Ly_Dryer  
P(Watts) = 1000  
I_Ly_Dryer = 4.167 A  
R_Ly_Dryer = 57.600  $\Omega$   
  
-----  
Name of Component: Ly_HP  
P(Watts) = 1000  
I_Ly_HP = 4.167 A  
R_Ly_HP = 57.600  $\Omega$   
-----
```

Figure 3.11: Laundry Calculations

3.3.12 For Garage

```
-----  
Name of Component: Ge_Bulb  
P(Watts) = 15  
I_Ge_Bulb = 0.062 A  
R_Ge_Bulb = 3840.000  $\Omega$   
-----  
Name of Component: Ge_Fan  
P(Watts) = 50  
I_Ge_Fan = 0.208 A  
R_Ge_Fan = 1152.000  $\Omega$   
-----  
Name of Component: Ge_Wpump  
P(Watts) = 600  
I_Ge_Wpump = 2.500 A  
R_Ge_Wpump = 96.000  $\Omega$   
-----  
Name of Component: Ge_HP  
P(Watts) = 1000  
I_Ge_HP = 4.167 A  
R_Ge_HP = 57.600  $\Omega$   
-----
```

Figure 3.12: Garage Calculations

3.4 Total Calculations

3.4.1 High Power Devices Calculations

```
>> H_I_sum  
  
H_I_sum =  
  
82.0000  
  
>> H_P_sum  
  
H_P_sum =  
  
19680
```

Figure 3.13: High Power Devices Calculations

3.4.2 Low Power Devices(UPS) Calculations

```
>> L_I_sum  
  
L_I_sum =  
  
6.5000  
  
>> L_P_sum  
  
L_P_sum =  
  
1.5600e+03
```

Figure 3.14: Low Power Devices(UPS) Calculations

3.4.3 Total Power Calculations

```
>> L_P_sum + H_P_sum  
  
ans =  
  
21240
```

Figure 3.15: Total Power Calculations

3.5 Results

1. The Main Power Supply should be at least 21.5kW.
2. The UPS Power should be at least 1.6kW.

Chapter 4

Discussion and Conclusion

4.1 Discussion

4.1.1 Load Estimation and Distribution

Accurate load estimation played a crucial role in ensuring that the residential electrical system could safely handle both high-power and low-power appliances. By grouping devices based on power ratings, the design reduced the risk of overload and allowed proper sizing of the main supply and UPS system.

4.1.2 Use of Analytical Tools (MATLAB)

MATLAB significantly improved the accuracy of calculating current, resistance, and total power for each room. These calculations ensured that every component operated within safe electrical limits and that the overall system capacity was properly defined.

4.1.3 Circuit Simulation and Validation (Multisim)

The circuit was validated using NI Multisim, which helped visualize the wiring layout, confirm current flow, and verify the behavior of protective devices. Color-coded wiring provided clarity and ensured compliance with safety standards during simulation.

4.1.4 Assumptions and Practical Considerations

The assumption of a 240 V DC supply simplified calculations, although real household systems use AC. Despite this simplification, the core engineering principles—including load management, safety, and wiring practices—remain valid and applicable to real-world scenarios.

4.2 Conclusion

This project successfully achieved its goal of designing and analyzing a complete residential electrical wiring system with an integrated UPS network. The final calculations show a required main supply capacity of 21.5 kW and a UPS capacity of 1.6 kW. The combined use of MATLAB for analytical computations and Multisim for simulation provided a reliable and accurate design process. Overall, the project demonstrates the importance of load distribution, safety devices, and systematic electrical planning, serving as a strong foundation for future enhancements such as smart systems or renewable energy integration.

Bibliography

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